

GATE EE Batch 005

Human Final QA - Mobile Fillable Review Packet

Human reviewer record

Name:

Role or qualification:

Review date (YYYY-MM-DD):

Reviewer identifier (optional):

Type the attestation exactly as printed below:

This packet contains 12 checksum-bound Batch 005 Electromagnetic Fields questions. Mathematical expressions are canonical LaTeX; declared SVG diagrams are checksum-bound and rendered below their stems.

Canonical source SHA-256: f30c6d8b1a522d75effdc46b147c87a723e57ecc0a43ef74624daa9da25fb89e

Strict Formatter report SHA-256: d1ed63b24f739b5df1b81ca035b3b52b5ecee81c56c6b72036143e12ec96dd8b

Automated prerequisite: **12 PASS / 0 REVIEW / 0 invalid**. This does not replace your human judgment.

Required procedure

1. Solve each question independently before relying on the declared answer.
2. Inspect every displayed diagram and all labels when a figure is present.
3. Select exactly one of PASS or FAIL for each of the five checks.
4. Select exactly one final decision: PASS, REVISE, or REJECT.
5. Explain every REVISE or REJECT decision in Notes.
6. Save this filled PDF and return it for checksum-bound import and validation.

Required attestation: I independently reviewed the Batch 005 questions, answers and solutions and approve only the question IDs marked PASS below.

State your actual background accurately in the qualification field; do not invent a title or credential.

TMB-GATE-EE-EMFT-001 - revision 1

Question 1 of 12

Route: Electrical Engineering -> Electromagnetic Fields -> Electrostatics

Type / marks / difficulty: MCQ / 1 / Easy

Question

A point charge Q is located at the centre of a closed spherical surface. Which value equals $\oint_S \mathbf{D} \cdot d\mathbf{S}$?

Options

- A. 0
- B. $Q/(4\pi)$
- C. Q
- D. Q/ϵ_0

Verification reference

Declared answer: C

Independent recomputation evidence: Gauss law for $\mathbf{D}$ makes the closed-surface flux equal to enclosed free charge Q .

Canonical solution:

Gauss law for electric flux density states $\oint_S \mathbf{D} \cdot d\mathbf{S} = Q_{\text{enclosed}}$. The sphere encloses the complete point charge, so the integral equals Q . Final answer: C

Human decision

Select exactly one box in each row. PASS for the item is valid only when all five checks are PASS.

Check	PASS	FAIL
Technical correctness	☐	☐
Answer correctness	☐	☐
Solution correctness	☐	☐
Clarity / ambiguity	☐	☐
Originality-conflict check	☐	☐

Decision (select exactly one): PASS ☐ REVISE ☐ REJECT ☐

Notes / independent calculation summary:

TMB-GATE-EE-EMFT-002 - revision 1

Question 2 of 12

Route: Electrical Engineering -> Electromagnetic Fields -> Electrostatics

Type / marks / difficulty: NAT / 1 / Easy

Question

An infinite line charge in free space has uniform density 55.6 nC/m . Determine the electric-field magnitude at radial distance 0.1 m ; enter the value in kV/m . Use $\epsilon_0 = 8.854 \times 10^{-12} \text{ F/m}$.

Verification reference

Declared answer: 9.994

Independent recomputation evidence: The line-charge field $\lambda/(2\pi\epsilon_0\rho)$ evaluates to 9.994 kV/m .

Canonical solution:

For an infinite line charge, $E = \frac{\lambda}{2\pi\epsilon_0\rho}$. Substitution gives $E = \frac{55.6 \times 10^{-9}}{2\pi(8.854 \times 10^{-12})(0.1)} = 9.994 \times 10^3 \text{ V/m} = 9.994 \text{ kV/m}$. Final answer: 9.994

Human decision

Select exactly one box in each row. PASS for the item is valid only when all five checks are PASS.

Check	PASS	FAIL
Technical correctness	☐	☐
Answer correctness	☐	☐
Solution correctness	☐	☐
Clarity / ambiguity	☐	☐
Originality-conflict check	☐	☐

Decision (select exactly one): PASS ☐ REVISE ☐ REJECT ☐

Notes / independent calculation summary:

TMB-GATE-EE-EMFT-003 - revision 1

Question 3 of 12

Route: Electrical Engineering -> Electromagnetic Fields -> Electrostatics

Type / marks / difficulty: MSQ / 2 / Hard

Question

The figure shows a charge-free boundary between dielectric 1 with $\epsilon_{r1} = 2$ and dielectric 2 with $\epsilon_{r2} = 5$. In medium 1, the electric field has normal component 3 V/m and tangential component 4 V/m. Select all correct statements for medium 2.

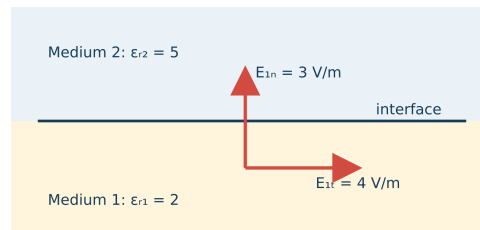


Figure: Charge-free dielectric interface

Diagram description: Horizontal dielectric interface with a normal axis, a tangential axis, and electric-field components in medium 1.

Options

- A. $D_{2n} = D_{1n}$
- B. $E_{2t} = E_{1t}$
- C. $E_{2n} = 1.2 \text{ V/m}$
- D. $|\mathbf{E}_2| = 5 \text{ V/m}$

Verification reference

Declared answer: A, B, C

Independent recomputation evidence: Charge-free boundary conditions preserve normal D and tangential E , giving $E_{2n} = 1.2 \text{ V/m}$.

Canonical solution:

With no free surface charge, the normal component of $\mathbf{D}$ is continuous, so $\epsilon_1 E_{1n} = \epsilon_2 E_{2n}$ and $E_{2n} = (2/5)(3) = 1.2 \text{ V/m}$. The tangential component of $\mathbf{E}$ is continuous, so $E_{2t} = 4 \text{ V/m}$. Therefore $|\mathbf{E}_2| = \sqrt{1.2^2 + 4^2} \neq 5 \text{ V/m}$. Final answer: A, B, C

Human decision

Select exactly one box in each row. PASS for the item is valid only when all five checks are PASS.

Check	PASS	FAIL
Technical correctness	☐	☐
Answer correctness	☐	☐
Solution correctness	☐	☐
Clarity / ambiguity	☐	☐
Originality-conflict check	☐	☐

Decision (select exactly one): PASS ☐ REVISE ☐ REJECT ☐

Notes / independent calculation summary:

TMB-GATE-EE-EMFT-004 - revision 1

Question 4 of 12

Route: Electrical Engineering -> Electromagnetic Fields -> Electrostatics

Type / marks / difficulty: MCQ / 1 / Medium

Question

Two concentric conducting spheres of radii a and b , with $b > a$, are separated by a dielectric of permittivity ε . Neglect fringing. Which expression gives the capacitance?

Options

- A. $4\pi\varepsilon(a + b)$
- B. $4\pi\varepsilon\frac{ab}{b-a}$
- C. $4\pi\varepsilon\frac{b-a}{ab}$
- D. $\frac{\varepsilon ab}{4\pi(b-a)}$

Verification reference

Declared answer: B

Independent recomputation evidence: Integrating the radial field between a and b gives $C = 4\pi\varepsilon ab/(b - a)$.

Canonical solution:

For charge Q , the potential difference is $V = \frac{Q}{4\pi\varepsilon}(\frac{1}{a} - \frac{1}{b}) = \frac{Q(b-a)}{4\pi\varepsilon ab}$. Thus $C = Q/V = 4\pi\varepsilon\frac{ab}{b-a}$. Final answer: B

Human decision

Select exactly one box in each row. PASS for the item is valid only when all five checks are PASS.

Check	PASS	FAIL
Technical correctness	☐	☐
Answer correctness	☐	☐
Solution correctness	☐	☐
Clarity / ambiguity	☐	☐
Originality-conflict check	☐	☐

Decision (select exactly one): PASS ☐ REVISE ☐ REJECT ☐

Notes / independent calculation summary:

TMB-GATE-EE-EMFT-005 - revision 1

Question 5 of 12

Route: Electrical Engineering -> Electromagnetic Fields -> Electrostatics

Type / marks / difficulty: NAT / 2 / Medium

Question

A solid sphere of radius 0.1 m carries a uniform volume charge with total charge 8 nC in free space. Determine the potential at the centre relative to the potential at the surface, in volts. Use $\epsilon_0 = 8.854 \times 10^{-12} \text{ F/m}$.

Verification reference

Declared answer: 359.5

Independent recomputation evidence: The centre-to-surface potential difference $Q/(8\pi\epsilon_0 a)$ evaluates to 359.5 V.

Canonical solution:

Inside a uniformly charged sphere, the centre-to-surface potential difference is $V(0) - V(a) = \frac{Q}{8\pi\epsilon_0 a}$.

Hence the requested value is $\frac{8 \times 10^{-9}}{8\pi(8.854 \times 10^{-12})(0.1)} = 359.5 \text{ V}$. Final answer: 359.5

Human decision

Select exactly one box in each row. PASS for the item is valid only when all five checks are PASS.

Check	PASS	FAIL
Technical correctness	☐	☐
Answer correctness	☐	☐
Solution correctness	☐	☐
Clarity / ambiguity	☐	☐
Originality-conflict check	☐	☐

Decision (select exactly one): PASS ☐ REVISE ☐ REJECT ☐

Notes / independent calculation summary:

TMB-GATE-EE-EMFT-006 - revision 1

Question 6 of 12

Route: Electrical Engineering -> Electromagnetic Fields -> Electrostatics

Type / marks / difficulty: NAT / 1 / Easy

Question

A parallel-plate capacitor has half of its plate area filled through the full separation with a dielectric of relative permittivity 3; the other half remains air. Neglect fringing. Enter the ratio of its capacitance to the all-air capacitance.

Verification reference

Declared answer: 2

Independent recomputation evidence: The air and dielectric half-area capacitances add in parallel to twice the all-air value.

Canonical solution:

The two half-area regions are in parallel. Thus $C = \frac{\varepsilon_0(A/2)}{d} + \frac{3\varepsilon_0(A/2)}{d} = 2\frac{\varepsilon_0 A}{d}$. Relative to the all-air value $\varepsilon_0 A/d$, the ratio is 2. Final answer: 2

Human decision

Select exactly one box in each row. PASS for the item is valid only when all five checks are PASS.

Check	PASS	FAIL
Technical correctness	☐	☐
Answer correctness	☐	☐
Solution correctness	☐	☐
Clarity / ambiguity	☐	☐
Originality-conflict check	☐	☐

Decision (select exactly one): PASS ☐ REVISE ☐ REJECT ☐

Notes / independent calculation summary:

TMB-GATE-EE-EMFT-007 - revision 1

Question 7 of 12

Route: Electrical Engineering -> Electromagnetic Fields -> Magnetostatics

Type / marks / difficulty: MCQ / 2 / Medium

Question

The figure shows two infinitely long parallel conductors separated by 0.2 m. The left conductor carries 5 A out of the page and the right conductor carries 10 A into the page. Determine the magnetic-flux-density magnitude at the midpoint.

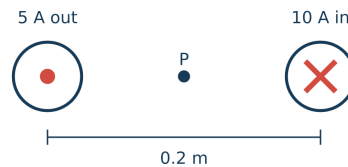


Figure: Parallel conductors and midpoint P

Diagram description: Two conductors 0.2 m apart, with 5 A out of the page on the left and 10 A into the page on the right; midpoint P is marked.

Options

- A. $10 \mu\text{T}$
- B. $15 \mu\text{T}$
- C. $30 \mu\text{T}$
- D. $45 \mu\text{T}$

Verification reference

Declared answer: C

Independent recomputation evidence: At the midpoint the two right-hand-rule fields add to $30 \mu\text{T}$.

Canonical solution:

The midpoint is 0.1 m from each conductor. By the right-hand rule the two fields point in the same direction there. Hence $B = \frac{\mu_0(5+10)}{2\pi(0.1)} = 30 \times 10^{-6} \text{ T} = 30 \mu\text{T}$. Final answer: C

Human decision

Select exactly one box in each row. PASS for the item is valid only when all five checks are PASS.

Check	PASS	FAIL
Technical correctness	☐	☐
Answer correctness	☐	☐
Solution correctness	☐	☐
Clarity / ambiguity	☐	☐
Originality-conflict check	☐	☐

Decision (select exactly one): PASS ☐ REVISE ☐ REJECT ☐

Notes / independent calculation summary:

TMB-GATE-EE-EMFT-008 - revision 1

Question 8 of 12

Route: Electrical Engineering -> Electromagnetic Fields -> Magnetostatics

Type / marks / difficulty: MSQ / 1 / Medium

Question

Select all statements that are valid in magnetostatics for ordinary media without magnetic monopoles.

Options

- A. $\oint_C \mathbf{H} \cdot d\mathbf{l}$ equals the free current enclosed by C .
- B. $\nabla \cdot \mathbf{B} = 0$.
- C. A free surface-current density can produce a discontinuity in tangential $\mathbf{H}$.
- D. The normal component of $\mathbf{B}$ may have an arbitrary discontinuity even when no magnetic monopoles exist.

Verification reference

Declared answer: A, B, C

Independent recomputation evidence: Ampere law, zero magnetic divergence and the surface-current jump in tangential H are valid; normal B is continuous.

Canonical solution:

Ampere law gives $\oint_C \mathbf{H} \cdot d\mathbf{l} = I_{\text{free, enclosed}}$. Gauss law for magnetism gives $\nabla \cdot \mathbf{B} = 0$, which also enforces continuity of normal $\mathbf{B}$. A free surface current produces the standard tangential- $\mathbf{H}$ jump. Therefore the first three statements are correct and the fourth is false. Final answer: A, B, C

Human decision

Select exactly one box in each row. PASS for the item is valid only when all five checks are PASS.

Check	PASS	FAIL
Technical correctness	☐	☐
Answer correctness	☐	☐
Solution correctness	☐	☐
Clarity / ambiguity	☐	☐
Originality-conflict check	☐	☐

Decision (select exactly one): PASS ☐ REVISE ☐ REJECT ☐

Notes / independent calculation summary:

TMB-GATE-EE-EMFT-009 - revision 1

Question 9 of 12

Route: Electrical Engineering -> Electromagnetic Fields -> Magnetic Circuits

Type / marks / difficulty: NAT / 2 / Hard

Question

A toroidal magnetic core has mean magnetic path length 0.5 m, relative permeability 1000, and a single air gap of length 1 mm. A 500-turn winding carries 0.2 A. Neglect leakage and fringing. Determine the flux density in tesla.

Verification reference

Declared answer: 0.0838

Independent recomputation evidence: The core and gap reluctance drops give $B = \mu_0 NI / (l_g + l_c / \mu_r) = 0.0838 \text{ T}$.

Canonical solution:

The same flux density crosses the core and gap. The magnetic-circuit relation is $NI = \frac{B}{\mu_0 \mu_r} l_c + \frac{B}{\mu_0} l_g$.

Thus $B = \frac{\mu_0 NI}{l_g + l_c / \mu_r} = \frac{(4\pi \times 10^{-7})(500)(0.2)}{0.001 + 0.5/1000} = 0.0838 \text{ T}$. Final answer: 0.0838

Human decision

Select exactly one box in each row. PASS for the item is valid only when all five checks are PASS.

Check	PASS	FAIL
Technical correctness	☐	☐
Answer correctness	☐	☐
Solution correctness	☐	☐
Clarity / ambiguity	☐	☐
Originality-conflict check	☐	☐

Decision (select exactly one): PASS ☐ REVISE ☐ REJECT ☐

Notes / independent calculation summary:

TMB-GATE-EE-EMFT-010 - revision 1

Question 10 of 12

Route: Electrical Engineering -> Electromagnetic Fields -> Electromagnetic Induction

Type / marks / difficulty: MCQ / 1 / Medium

Question

A single-turn loop links magnetic flux $\Phi(t) = (4t^2 + 2t)$ mWb. Determine the magnitude of the induced emf at $t = 0.5$ s.

Options

- A. 2 mV
- B. 4 mV
- C. 6 mV
- D. 8 mV

Verification reference

Declared answer: C

Independent recomputation evidence: Differentiating $(4t^2 + 2t)$ mWb at $t = 0.5$ s gives 6 mV.

Canonical solution:

Faraday law gives $|e| = |d\Phi/dt|$. Here $d\Phi/dt = (8t + 2)$ mWb/s, so at $t = 0.5$ s the magnitude is 6 mV.
Final answer: C

Human decision

Select exactly one box in each row. PASS for the item is valid only when all five checks are PASS.

Check	PASS	FAIL
Technical correctness	☐	☐
Answer correctness	☐	☐
Solution correctness	☐	☐
Clarity / ambiguity	☐	☐
Originality-conflict check	☐	☐

Decision (select exactly one): PASS ☐ REVISE ☐ REJECT ☐

Notes / independent calculation summary:

TMB-GATE-EE-EMFT-011 - revision 1

Question 11 of 12

Route: Electrical Engineering -> Electromagnetic Fields -> Electromagnetic Induction

Type / marks / difficulty: NAT / 2 / Medium

Question

A conducting rod of length 0.5 m moves at 4 m/s perpendicular to a uniform magnetic field of 0.8 T. The rod completes a circuit of total resistance $2\ \Omega$. Neglect friction. Enter the external force required to maintain constant speed, in millinewtons.

Verification reference

Declared answer: 320

Independent recomputation evidence: Motional emf is 1.6 V, current is 0.8 A and $BIl = 0.32\ \text{N} = 320\ \text{mN}$.

Canonical solution:

The motional emf is $e = Blv = (0.8)(0.5)(4) = 1.6\ \text{V}$, so $I = e/R = 0.8\ \text{A}$. The opposing magnetic force is $F = BIl = (0.8)(0.8)(0.5) = 0.32\ \text{N} = 320\ \text{mN}$. Final answer: 320

Human decision

Select exactly one box in each row. PASS for the item is valid only when all five checks are PASS.

Check	PASS	FAIL
Technical correctness	☐	☐
Answer correctness	☐	☐
Solution correctness	☐	☐
Clarity / ambiguity	☐	☐
Originality-conflict check	☐	☐

Decision (select exactly one): PASS ☐ REVISE ☐ REJECT ☐

Notes / independent calculation summary:

TMB-GATE-EE-EMFT-012 - revision 1

Question 12 of 12

Route: Electrical Engineering -> Electromagnetic Fields -> Electromagnetic Induction

Type / marks / difficulty: MCQ / 2 / Hard

Question

The cross-sectional figure shows a long solenoid of radius 0.1 m whose uniform internal magnetic flux density changes at $dB/dt = 2 \text{ T/s}$. Neglect the field outside the solenoid. Determine the induced electric-field magnitude on a circular path of radius 0.2 m concentric with the solenoid.

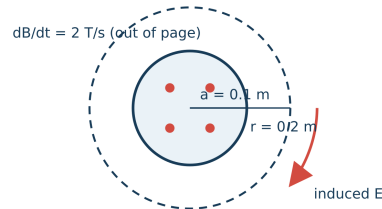


Figure: Solenoid cross-section and circular observation path

Diagram description: Concentric circles of radii 0.1 m and 0.2 m representing the solenoid cross-section and the integration path, with increasing axial magnetic field.

Options

- A. 0.0125 V/m
- B. 0.025 V/m
- C. 0.05 V/m
- D. 0.10 V/m

Verification reference

Declared answer: C

Independent recomputation evidence: For $r > a$, Faraday law gives $E = a^2 |dB/dt| / (2r) = 0.05 \text{ V/m}$.

Canonical solution:

For $r > a$, Faraday law gives $2\pi r E = \pi a^2 |dB/dt|$. Therefore $E = \frac{a^2}{2r} |dB/dt| = \frac{0.1^2}{2(0.2)} (2) = 0.05 \text{ V/m}$.
Final answer: C

Human decision

Select exactly one box in each row. PASS for the item is valid only when all five checks are PASS.

Check	PASS	FAIL
Technical correctness	☐	☐
Answer correctness	☐	☐
Solution correctness	☐	☐
Clarity / ambiguity	☐	☐
Originality-conflict check	☐	☐

Decision (select exactly one): PASS ☐ REVISE ☐ REJECT ☐

Notes / independent calculation summary:

Review completion

Complete this page only after reviewing all 12 questions.

Final reviewed-results decision: APPROVE REVIEWED RESULTS ☐

Reviewer typed signature:

Final comments:

Paper eligibility remains blocked until this completed record is validated and only explicitly approved IDs are promoted.